FINITELY MANY TAYLOR COEFFICIENTS OF A FORMAL LOCAL MAP ARE CONTROLLED BY ONE FINITE TYPE ETALE-LIKE MODIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is the whole lemma environment lines 11529-11542; proof span is lines 11544-11568, the immediately following proof environment.

Editorial note. The proof cites no label and no bibliography entry, so no external_references are needed and the assembler adds no navigation stubs.

Editorial note. The proof text is retained verbatim, including the original's use of the same symbol sigma for the map and for an arbitrary competitor map; no notational repair was applied.

Editorial note. Difficulty H3 basis: the simultaneous control of all factorisations modulo m_s^N is a genuine research-level finiteness construction, not a routine consequence of the definitions.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Noetherian local rings and their completions, finite type ring maps and their presentations by generators and relations, principal localisations, and the Taylor expansion of an element of the completion along a generating sequence of the maximal ideal. The control construction is retained in full.

Lemma 0.1. *Let $T \to S$ be finite type morphisms of Noetherian schemes. Let $t \in T$ map to $s \in S$ and let $\sigma : \mathcal{O}_{T,t} \to \mathcal{O}^\wedge_{S,s}$ be a local $\mathcal{O}_{S,s}$-algebra map. For every $N \geq 1$ there exists a finite type morphism $(T', t') \to (T, t)$ such that σ factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$ and such that for every local $\mathcal{O}_{S,s}$-algebra map $\sigma' : \mathcal{O}_{T,t} \to \mathcal{O}^\wedge_{S,s}$ which factors through $\mathcal{O}_{T,t} \to \mathcal{O}_{T',t'}$ the maps σ and σ' agree modulo $\mathfrak{m}_s^N$.*

Proof. We may assume S and T are affine. Say $S = \mathrm{Spec}(R)$ and $T = \mathrm{Spec}(C)$. Let $c_1, \ldots, c_n \in C$ be generators of C as an R-algebra. Let $\mathfrak{p} \subset R$ be the prime ideal corresponding to s. Say $\mathfrak{p} = (f_1, \ldots, f_m)$. After replacing R by a principal localization (to clear denominators in $R_\mathfrak{p}$) we may assume there exist $r_1, \ldots, r_n \in R$ and $a_{i,I} \in \mathcal{O}^\wedge_{S,s}$ where $I = (i_1, \ldots, i_m)$ with $\sum i_j = N$ such that

$$\sigma(c_i) = r_i + \sum\nolimits_I a_{i,I} f_1^{i_1} \ldots f_m^{i_m}$$

in $\mathcal{O}^\wedge_{S,s}$. Then we consider

$$C' = C[t_{i,I}] / \left(c_i - r_i - \sum\nolimits_I t_{i,I} f_1^{i_1} \ldots f_m^{i_m} \right)$$

with $\mathfrak{p}' = \mathfrak{p}C' + (t_{i,I})$ and factorization of $\sigma : C \to \mathcal{O}^\wedge_{S,s}$ through C' given by sending $t_{i,I}$ to $a_{i,I}$. Taking $T' = \mathrm{Spec}(C')$ works because any σ' as in the statement of the lemma will send c_i to r_i modulo the maximal ideal to the power N. $\square$

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